HAWK

HOSTILE ASSESSMENT WORKIT

An open-source pentest control panel for Kali Linux.
One window. Eight kill-chain phases. Every tool you already use.

VERSION 1.0.0

first public release · more to come

CREATED BY

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AUTHORIZED USE ONLY

HAWK is a tool for licensed penetration testers, CTF players, and red-teamers.
Never run it against systems you don't own or have explicit written permission to test.

Apache 2.0 · Open source · Attribution required

WHAT IS HAWK?

HAWK is a desktop GUI that wraps the standard Kali Linux pentesting toolkit into a single coherent workflow. Instead of juggling a dozen terminal tabs, you click through eight kill-chain phases and HAWK runs the right tool with the right arguments — then captures the results into structured findings.

EIGHT PHASES

RECON	passive footprinting — whois, dnsrecon, theHarvester, subfinder
SCANNING	port discovery — nmap, masscan, rustscan
ENUMERATION	service drill-down — gobuster, ffuf, enum4linux, kerbrute, ldapsearch
VULN SCAN	find weak spots — nikto, nuclei, wpscan, sqlmap, searchsploit
EXPLOITATION	land a shell — metasploit, hydra, responder, impacket, certipy
POST-EXPLOIT	live off the land — find SUID, sudo -l, scheduled tasks
LATERAL	pivot through — chisel, ligolo, proxychains, ssh tunnels
REPORTING	what you actually found, exportable

THE BIG IDEA

HAWK does not replace your tools. It orchestrates them.
Every command is visible, every flag is editable, every result is logged.
When you outgrow the GUI, the underlying CLI workflow is right there —
no proprietary lock-in, no hidden magic.

WHY HAWK EXISTS

Every CTF player and junior pentester hits the same wall: too many tools, too many flags, too much context-switching. The senior on the team has muscle memory; the junior has 12 browser tabs of cheat sheets.

HAWK was built to flatten that learning curve without dumbing anything down. It's the assistant that whispers "try gobuster first", "this port is SMB, use enum4linux", "that hash looks like NTLMv2" — without taking the wheel away from you.

WHY IT FEELS EASY

Guided dialogs	Every tool with non-obvious args (sqlmap, nuclei, certipy, hydra) opens a dialog that explains mode suggests parameters, and shows the live command preview before launch.
Credential Store	Cracked passwords, NTLM hashes, Kerberos tickets — all in one place. Other tools auto-pull from it: no copy-paste between terminals.
Target Intel	As scans complete, results are parsed into structured findings — open ports, services, versions, CVEs — searchable and exportable.
Background jobs	Long-running scans (hydra, nuclei, nmap full) keep going while you work on other phases. Each shows elapsed + estimated time.
Smart errors	Tool crashed? HAWK explains why (missing dep, port conflict, wrong wordlist) and offers a one-click fix.

DESIGN PRINCIPLE

"A junior pentester should land their first shell with HAWK and learn the underlying tools by reading what HAWK runs."

WHAT'S INSIDE

HAWK is ~15 000 lines of Python 3.11+ on top of customtkinter. It ships as source — clone, install deps, run. No daemon, no database.

STACK

Language	Python 3.11+
GUI	customtkinter (tk-based)
Runner	subprocess + threading
Sessions	JSON on disk
Reports	HTML + Markdown export
Platforms	Kali Linux (primary)
	Parrot, Ubuntu (tested)
License	Apache 2.0

INTEGRATED TOOLS

Recon	whois dig dnsrecon theHarvester
	subfinder amass whatweb wafw00f
Scanning	nmap masscan rustscan
Enum	gobuster ffuf enum4linux netexec
	kerbrute ldapsearch smbclient
Vuln	nikto nuclei wpscan sqlmap
	searchsploit CVE/NVD lookup
Exploit	metasploit hydra responder
	impacket-* certipy-ad
Lateral	chisel ligolo proxychains
Misc	john hashcat evil-winrm

ENGINEERING HIGHLIGHTS

- Cross-distro tool detection — finds binaries even when not on $PATH
- Auto-fix dialog for Python 3.12+ distutils crashes (apt impacket)
- Live command preview in every dialog before you hit RUN
- Background job bar with CPU-load warning when too many run at once
- Autosave every 5 min — no lost sessions on crashes
- Right-click context menus, global mouse-wheel scrolling, dark theme
- ANSI banner stripping — only real findings reach the Intel panel

THIS IS v1.0.0

First public release. The backend is solid — battle-tested across HTB, THM, and live engagements. What's listed below is what ships today; what comes next will be shaped by you, the people actually using HAWK.

V1.0.0 — WHAT SHIPS TODAY

- 8-phase kill-chain GUI
- 30+ integrated Kali tools
- Guided dialogs for every non-trivial tool
- Credential Store with auto-propagation
- Target Intel — structured finding parser
- Background jobs + CPU-load warnings
- Session save / load / 5-min autosave
- Smart error dialogs with one-click fixes

Apache 2.0 · attribution required · Kali Linux primary target

WHAT COMES NEXT

The roadmap is intentionally blank. HAWK v1.0.0 is a complete tool on its own — it does what it says on the cover. Where it goes from here will be driven by real feedback from real engagements.

Use it. Break it. File an issue. Open a PR. That's the roadmap.

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GitHub: to be announced · License: Apache 2.0 · Contact: via repo issues